You may be wondering why the x-axis ranges from 0-3 and the y-axis from 1-4. If you provide a single list or array to the plot () command, matplotlib assumes it is a sequence of y values, and automatically generates the x values for you. Since python ranges start with 0, the default x vector has the same length as y but starts with 0. Hence the x data are [0, 1, 2, 3].

plot() is a versatile command, and will take an arbitrary number of arguments.

Program

For example, to plot x and y, you can issue the command:

```
import matplotlib.pyplot as plt
plt.plot([1,2,3,4], [1,4,9,16])
plt.show()
```

This .plot takes many arguments, but the first two here are 'x' and 'y' coordinates. This means, you have 4 co-ordinates according to these lists: (1,1), (2,4), (3,9) and (4,16).

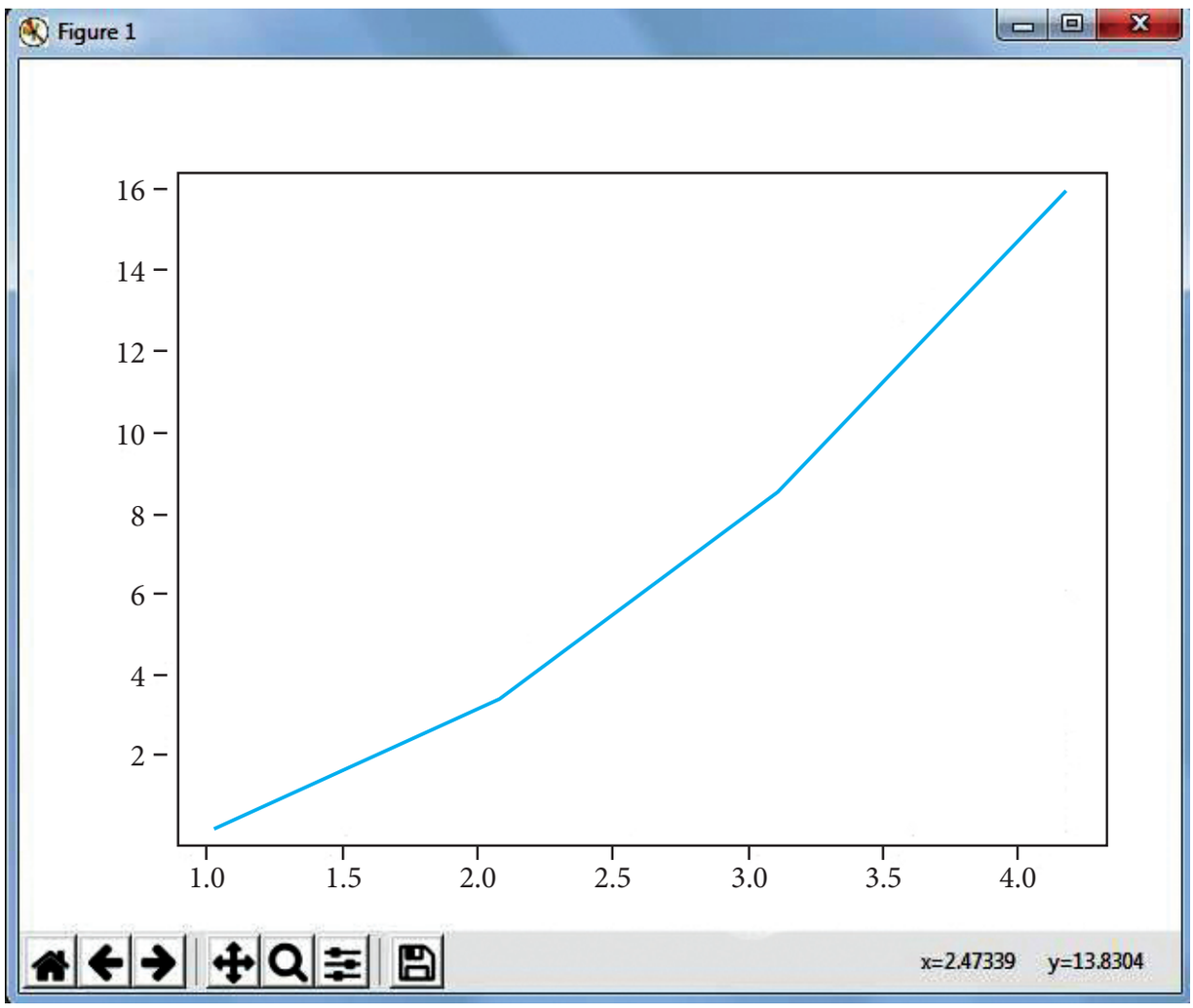


Figure 16.2

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